In while loop, the test expression is evaluated and if the test expression result is true, then the body of the loop is executed and again the control is transferred to the while loop. When the test expression result is false the control is transferred to statement-x.

Illustration 10.10 C++ program to sum numbers from 1 to 10 using while loop

```
#include <iostream>
using namespace std;
int main ()
{
int i=1,sum=0;
while(i<=10)
{
        sum=sum+i;
        i++;
}
cout<<"The sum of 1 to 10 is "<<sum;
return 0;
}
```

Output

```
The sum of 1 to 10 is 55
```

In the above program, the integer variable i is initialized to 1 and the variable sum to 0. The while loop checks the condition, i < 10, if the condition is true, the value of i, which is added to sum and i is incremented by 1. Again, the condition i < 10 is checked. Since 2 < 10, 2 is added to the earlier value of sum. This continues until i becomes 11. At this point in time, 11 < 10 evaluates to false and the while loop terminates. After the loop termination, the value of sum is displayed.

Illustration 10.11 C++ program to find sum and average of 5 numbers using while loop

```
#include <iostream>
using namespace std;
int main ()
{
int i=1,num,avg,sum=0;
while(i<=5)
{
        cout<<"Enter the number : ";
        cin>>num;
        sum=sum+num;
        i++;
}
avg=sum/5;
cout<<"The sum is "<<sum<<endl;
cout<<"The average is "<<avg;
return 0;
}
```

Output

```
Enter the number : 1
Enter the number : 2
Enter the number : 3
Enter the number : 4
Enter the number : 5
The sum is 15
The average is 3
```


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